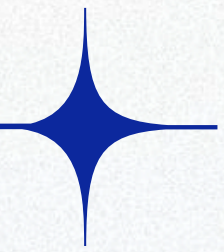




Air Quality Analysis (PM2.5) in Selected Countries – 2023

A Comparative Statistical Study

Ummuhan OZ/ 153246



Overview

The aim of this study is to analyze and compare PM2.5 air pollution levels across selected countries and to investigate whether there is a significant relationship between air quality and economic development. This helps to understand how economic factors may influence environmental conditions in different regions.

Country	PM2.5 ($\mu\text{g}/\text{m}^3$)	GDP per capita
Bangladesh	79,9	1.568
Pakistan	73,7	1.568
India	60	2.480
China	35	10.430
Poland	20	19.000
Turkey	18	12.986
Australia	5	67.979
Estonia	4,5	32.000
New Zealand	4	48.234

The dataset contains average PM2.5 (fine particulate matter) concentrations for nine countries in the year 2023.

These values are measured in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$).

PM2.5 is one of the most harmful air pollutants due to its ability to penetrate deep into the lungs and bloodstream.

The countries selected represent a range of economic and environmental conditions across different continents.

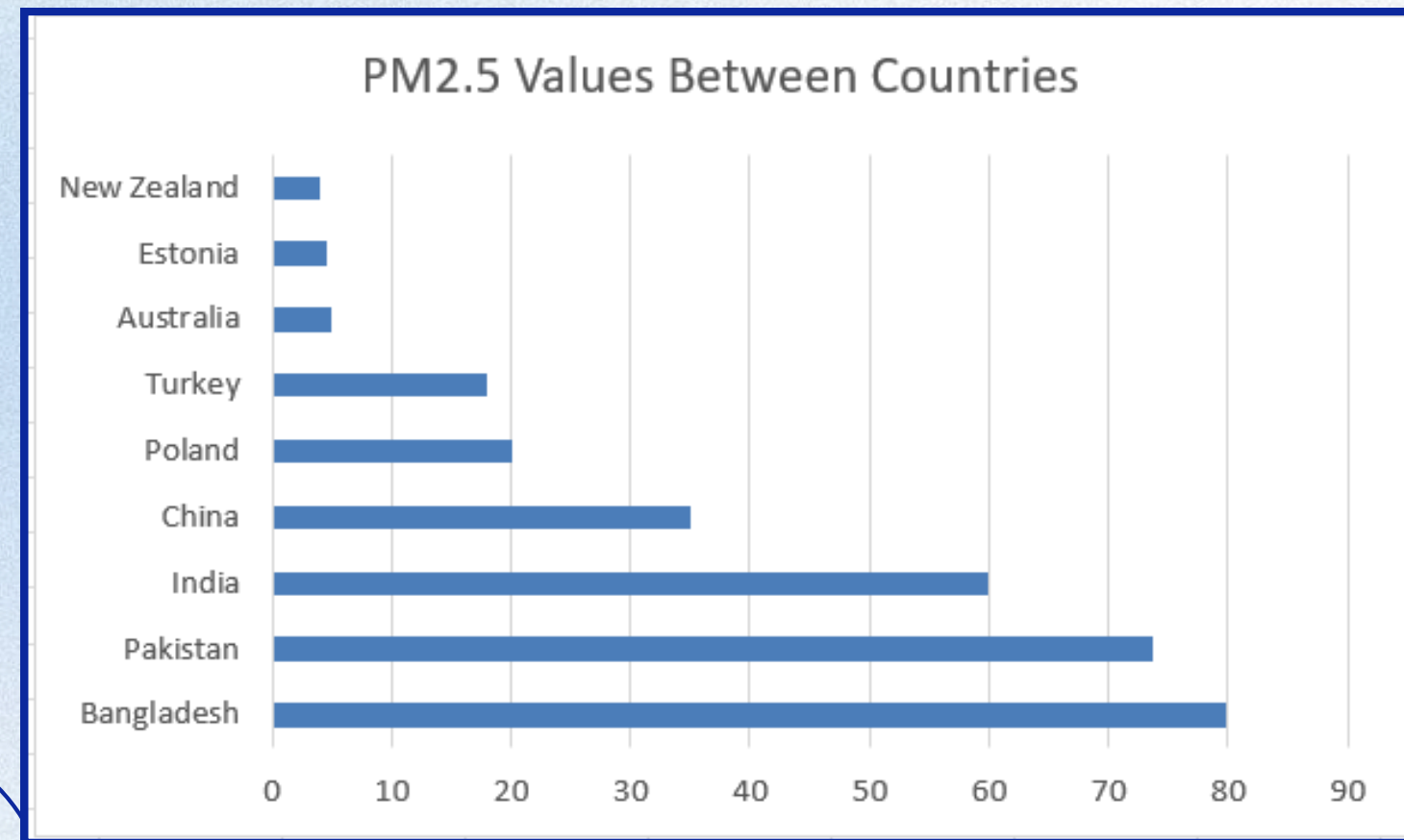
Descriptive Analysis of PM2.5 Levels

The descriptive analysis of PM2.5 concentrations across nine countries provides insights into the general distribution of air pollution levels.

- Mean: The average PM2.5 level is approximately 33.9 $\mu\text{g}/\text{m}^3$.
- Median: The middle value is 20 $\mu\text{g}/\text{m}^3$, indicating a right-skewed distribution.
- Standard Deviation: About 29.4, showing large variability among countries.
- Minimum and Maximum: PM2.5 values range from 4 $\mu\text{g}/\text{m}^3$ (New Zealand) to 79.9 $\mu\text{g}/\text{m}^3$ (Bangladesh).
- These statistics suggest significant differences in air quality between developing and developed countries.

mean	33,34	
median	20	
Standart D	28,71	
max	79,9	(Bangladesh)
min	4	(New Zealand)
count	9	

PM2.5 Levels by Country (2023)



This bar chart visually compares PM2.5 levels across the selected countries.

Bangladesh, Pakistan, and India show the highest levels of fine particulate pollution, well above the global safety standards.

In contrast, countries like Australia, Estonia, and New Zealand maintain much cleaner air.

The large disparity highlights global inequalities in environmental conditions, often linked to economic and regulatory differences.

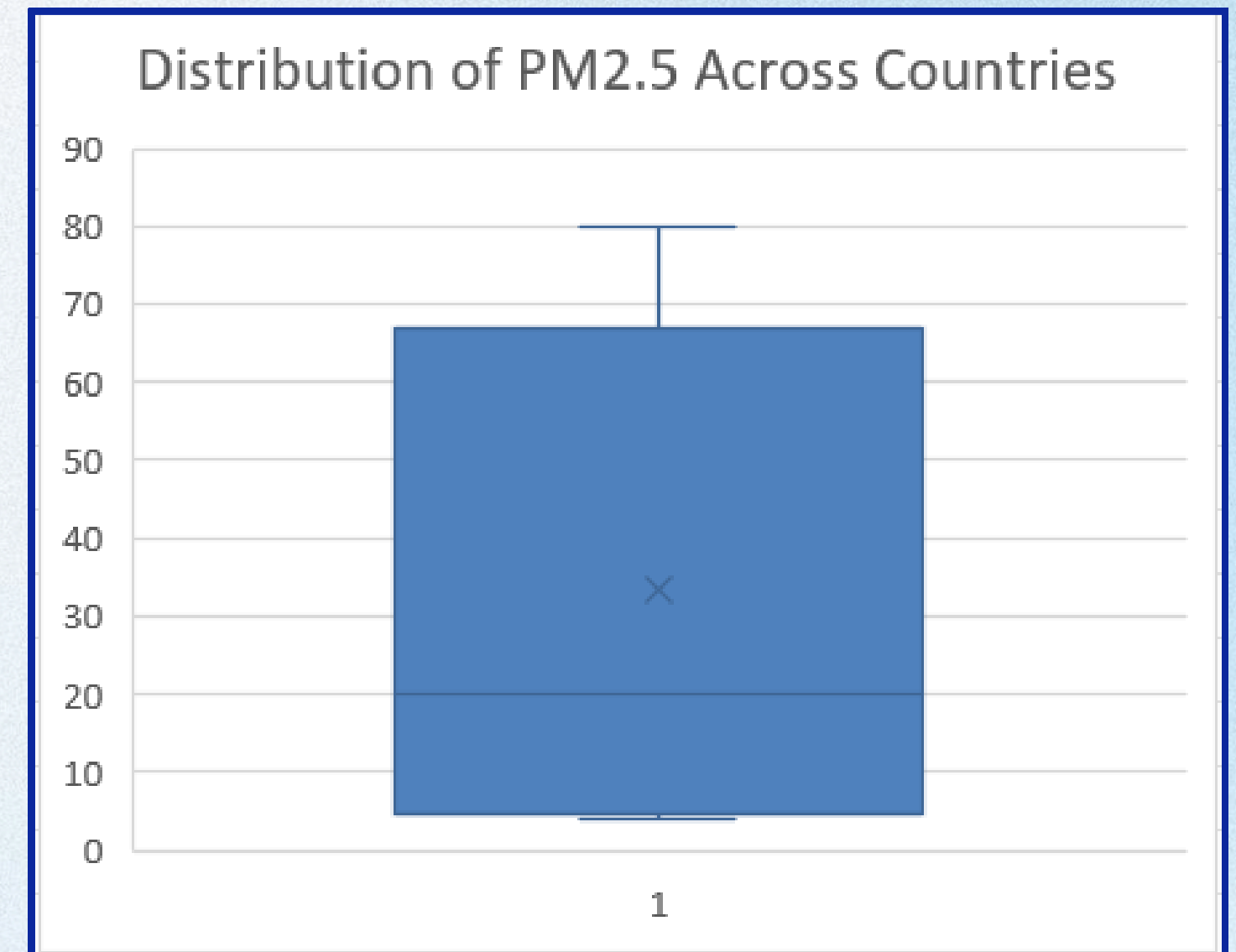
Boxplot of PM2.5 Concentrations

The boxplot illustrates the distribution and spread of PM2.5 levels across the selected countries.

The plot clearly shows a wide interquartile range, indicating a high level of variability.

Several countries fall into the upper whisker or above the upper quartile, representing outliers with extreme pollution, such as Bangladesh and Pakistan.

Countries with cleaner air are clustered near the lower quartile, indicating better air quality and less variability in those regions.



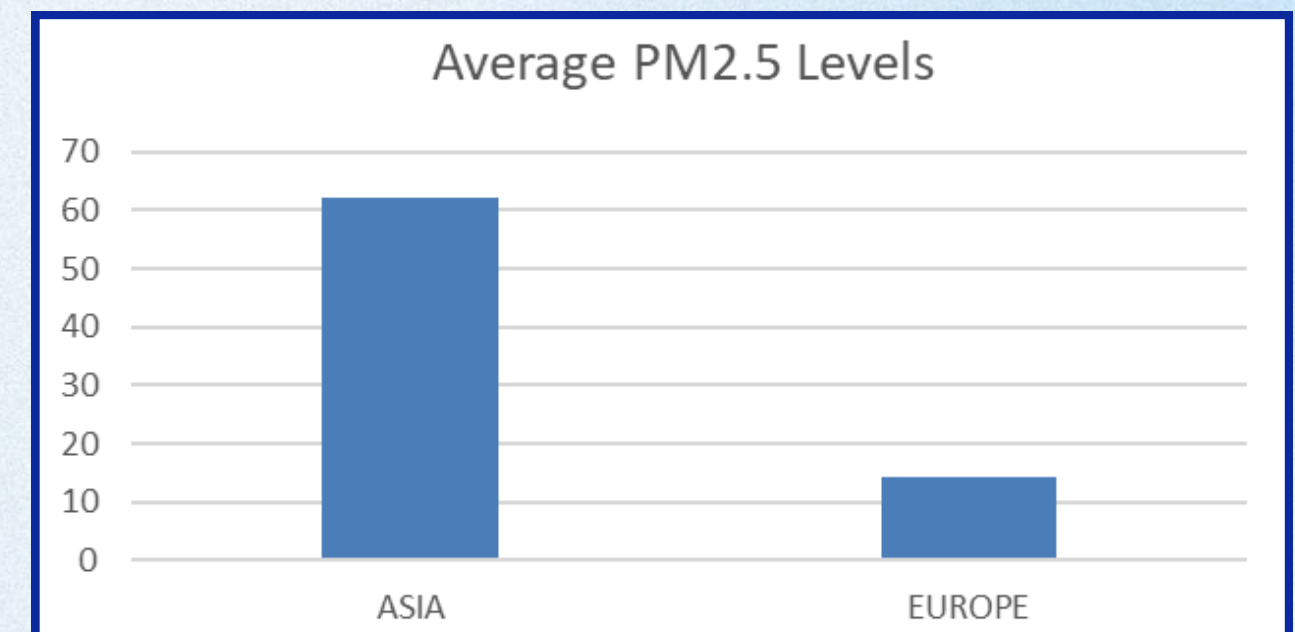
Regional Differences in Air Pollution

Based on a two-sample t-test assuming unequal variances, we tested whether there is a significant difference in PM2.5 levels between Asian and European countries.

The p-value from the test was 0.012, which is less than 0.05, so we reject the null hypothesis.

There is a statistically significant difference between the regions. Asian countries have significantly higher average PM2.5 levels than European countries.

	ASIA	EUROPE
Mean	62,15	14,16667
Variance	396,7367	71,08333
Observations	4	3
Hypothesized Mean Difference	0	
df	4	
t Stat	4,328648	
P(T<=t) one-tail	0,006181	
t Critical one-tail	2,131847	
P(T<=t) two-tail	0,012363	
t Critical two-tail	2,776445	

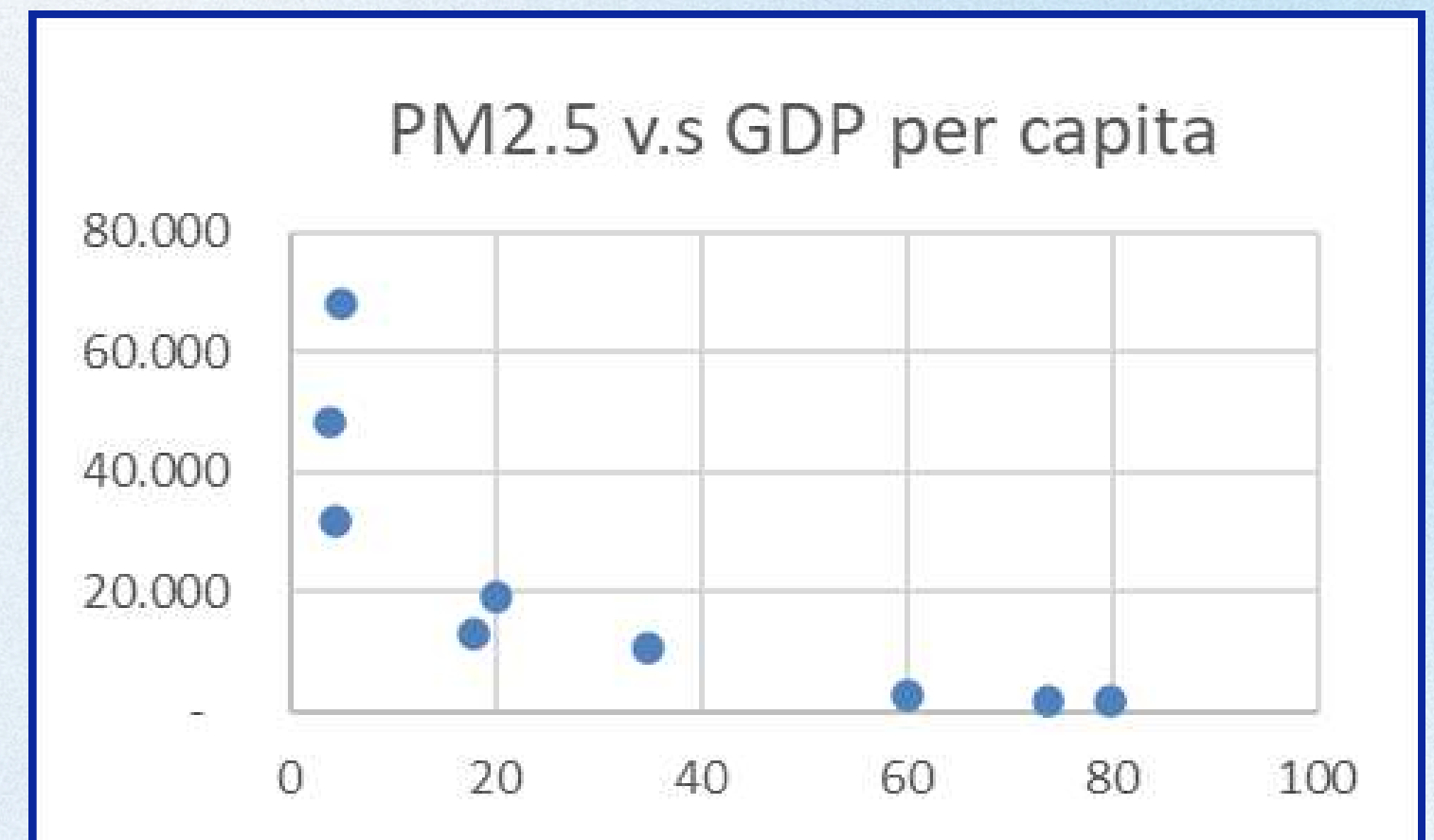


Correlation Between PM2.5 Levels and GDP per Capita

The Pearson correlation coefficient between PM2.5 concentrations and GDP per capita across the nine countries is -0.79, indicating a strong negative relationship.

This means that countries with higher GDP per capita tend to have significantly lower levels of air pollution (PM2.5).

In other words, economic development appears to be associated with improved air quality, likely due to better environmental regulations, cleaner technologies, and higher public awareness in wealthier nations.



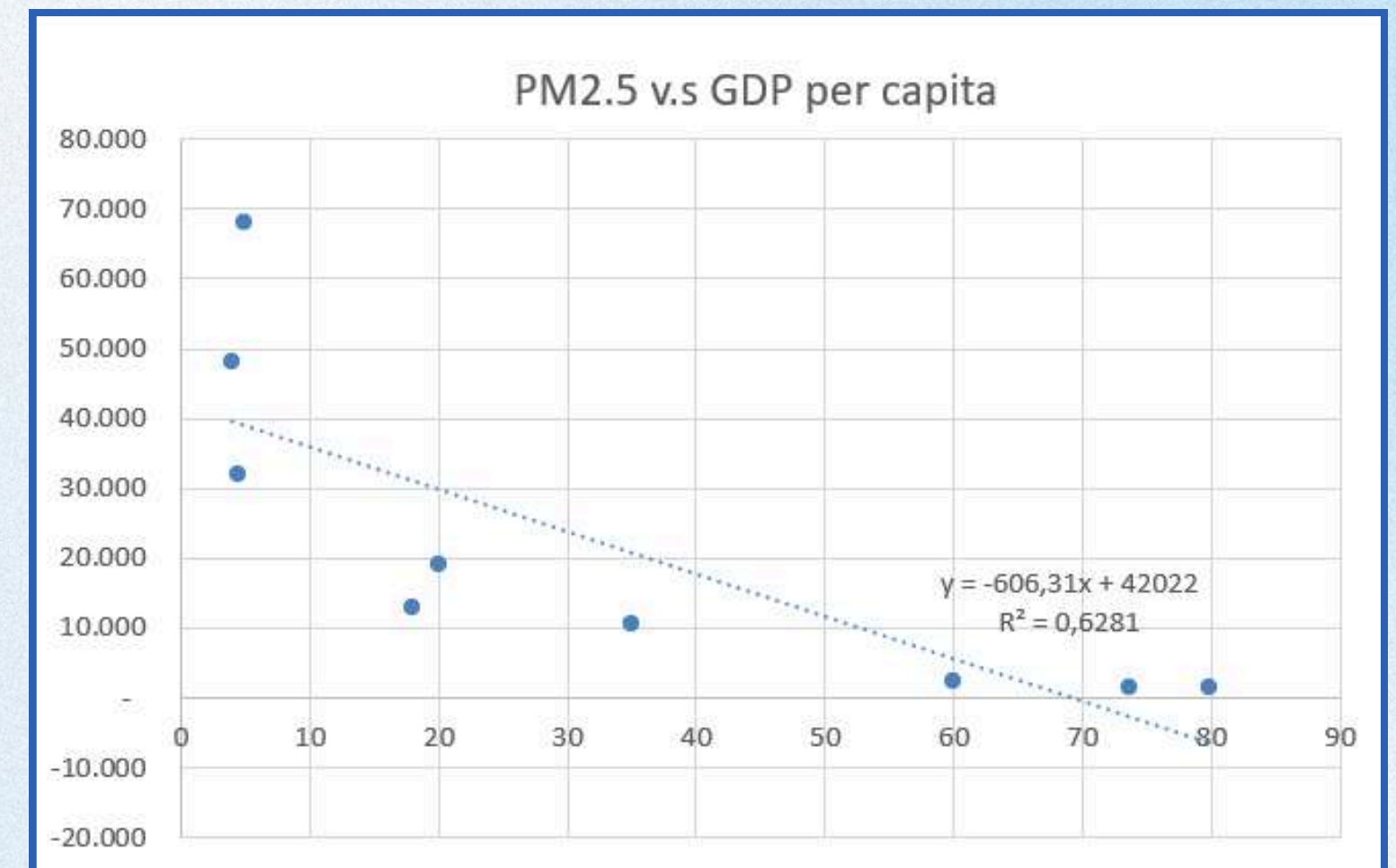
Linear Regression of PM2.5 on GDP per Capita

The linear regression model explains approximately 63% ($R^2 = 0.63$) of the variation in PM2.5 levels based on GDP per capita.

The negative slope indicates that as GDP per capita increases, PM2.5 pollution decreases significantly.

This suggests that economic growth plays a crucial role in reducing air pollution, possibly through investments in cleaner energy sources, pollution control measures, and urban planning.

These findings highlight the importance of economic development alongside environmental policies to improve air quality.



Key Findings

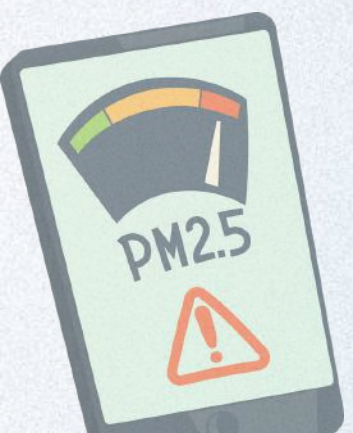
PM2.5 levels vary greatly across countries, from 4 $\mu\text{g}/\text{m}^3$ (New Zealand) to 79.9 $\mu\text{g}/\text{m}^3$ (Bangladesh).

Asian countries show significantly higher air pollution compared to non-Asian countries (t-test $p = 0.012$).

There is a strong negative correlation between PM2.5 and GDP per capita ($r = -0.79$).

Regression analysis confirms that economic development explains about 63% of the variation in PM2.5 levels.

These results suggest that both regional context and economic factors play critical roles in air quality.



Conclusion

This statistical analysis reveals clear patterns in global air quality:

Wealthier countries tend to have cleaner air, likely due to better infrastructure and environmental policies.

Significant regional differences highlight the need for global cooperation in pollution reduction efforts.

Data sources:

<https://www.iqair.com/world-air-quality-report>

<https://data.worldbank.org/indicator/NY.GDP.PCAP.CD>